simply means that we are trying to update motorcycles according to the temperature and other conditions they are deployed in." For the record, Raptee was once a partner to electric bus OEMs. However, it now wants to concentrate on being an OEM. Arjun highlighted that Raptee will never supply to a direct competitor!

The startup chose high voltage drivetrains because it feels the same is necessary to help India and Indians transition from ICE (internal combustion engine) to EV (electric vehicle). According to Arjun, a lot of E2Ws available in India today cannot do steep slopes without the motor getting overheated. A high voltage drivetrain, he highlights, can do the same better than an ICE vehicle without getting the motor heated.

"You cannot request a Toyota Innova owner to switch to a 1000cc CNG (compressed natural gas) powered car by telling him that it is good for his pocket as well as for the environment. You will have to give him something equivalent to that Innova experience," he says.

Cost is not a challenge for Raptee because of the indigenisation and vertical integration it has been able to achieve so far. Arjun adds that as most of the 250cc machines in India are available in the price range upwards of ₹200,000, Raptee's first electric motorcycle will be positioned in the same category. The startup will start retailing the same next year in Bengaluru and Chennai. It is targeting to open six experience centres in six cities before the end of the next year, each owned and operated by itself.

In 2025, the startup plans to introduce its second offering, targeted towards the 150cc motorcycle market, and aims to open at least 50 centres via the dealership route. Arjun says Raptee chose its first product as an electric motorcycle because motorcycles are the biggest medium of transport and people movement in India. According to him, most of the players in the competition will not transition to high voltage powertrains because the ecosystem of components is already established for them, and doing the same will require them to start from scratch.

Raptee's first product will feature a battery close to 6kW, which will promise an IDC (Indian driving cycle) range of 220km and a real-world range of close to 160km. It has so far raised ₹370 million from Director of Shankara Building Products, Eugene Mayne, Founder and CEO of Tristar Global (UAE), Lakshmi Narayanan, Former CEO of Cognizant, Ramesh Kannan, MD of Kaynes Technology, Chandu Nair, and other HNIs (high net worth individuals) from India and the UAE.

—Mukul Yudhveer Singh

2 HARNESSING THE SHAKTI FOR MICROCONTROLLER CHIPS

"As an academic institution, we've done everything we can to develop the Shakti microprocessor, and now I need you to make it accessible to the world" — Prof. Kamakoti Veezhinathan

It was a momentous occasion when IIT Madras announced the release of the Shakti microprocessor in 2018. While the adoption of this chip across strategic sectors, such as defence, nuclear power installations, government agencies and departments, is yet to be seen on a massive scale, startups such as Mindgrove Technologies are working towards making it accessible to the world.

Founded and incubated under IIT Madras in 2021 by co-founders and former colleagues, Shashwath TR (CEO) and Sharan Srinivas J (CTO), fabless semiconductor startup Mindgrove Technologies harnesses the potential of open source technology and the RISC-V Instruction Set Architecture (ISA) to create a high-performance microcontroller chip operating on a 28-nanometre technology node with a SHAKTI core.

The Shakti processor's open source nature, released under a permissive BSD three-clause license, eliminates licensing fees and sales royalties typically associated with buying proprietary chips, reducing costs. "The RISC-V ISA combines lessons from legacy ISAs like x86, ARM, MIPS, and more, resulting in impressive performance efficiency for resources invested. The chip's microarchitecture is designed to be simple, minimising area and power consumption, thus reducing production costs linked to chip size. We benefit from financial and infrastructure support and mentorship provided by IIT Madras, helping us identify and address potential issues early on," says Shashwath.

The choice of a 28-nanometre technology node, however unconventional, allows the company de-